Nicolás Maire Bravo



Computer Science Student · Software Developer · AI & Cybersecurity

Madrid, Spain — open to on-site or hybrid in Madrid, or fully remote worldwide.

Professional summary

Computer Science student at UC3M with exchange semesters at the Università di Bologna (Italy) and UADE (Argentina), focused on Artificial Intelligence, Cybersecurity and full-stack/backend development. I ship end-to-end products, from machine-learning models and secure APIs to distributed systems and automation. Fast learner, strong team player and most at home solving low-level or complex architectural problems.

Technical skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL, R, HTML/CSS
AI & Data: PyTorch, Transformers, Hugging Face, LLMs, RAG, Deep Learning, NumPy/Pandas, Jupyter
Backend & Web: FastAPI, React, Next.js, Supabase, PostgreSQL, REST APIs, n8n
Cybersecurity: Prompt-injection mitigation, AES-GCM, envelope encryption (KMS/DEK/KEK), network security
Systems & DevOps: Linux, Docker, Git, distributed systems, TCP/UDP sockets, POSIX, MQTT, Raspberry Pi

Languages

Spanish — native
English — C1 Advanced (Cambridge)
Italian — advanced
French — intermediate
Chinese — HSK3→HSK5

Contact

nico.maibra@gmail.com

linkedin.com/in/nicolás-maire-bravo

github.com/nico-maire

nico-maire.github.io

Experience

CitaSalon.online — B2B SaaS *2026 – Present*
Founder & Full-Stack Developer

- Founded and built a multi-tenant B2B SaaS platform (Next.js, FastAPI, Supabase/PostgreSQL) that digitises salons and local businesses: online booking, appointment management and automated notifications.

VIA · VisiónIAutomatización *2025 – 2026*
Founder, Full-Stack Developer & Automation Architect

- Founded and built a full-stack "VIA" automation platform (React/FastAPI) and backend services that connect n8n workflows with custom Python scripts and Hugging Face models.

AI Agent Automation Builder *2025*
Personal project — Python, AI agents

- Built a tool that uses interconnected AI agents to analyse business logic and auto-generate complex automation workflows.

Selected projects

Neural Calculator & Generalisation Engine · Python, PyTorch, Transformers, Deep Learning
Trained Seq2Seq and Transformer models at UniBO and measured in-distribution vs. out-of-distribution and length generalisation.
LLM Security & RAG Encryption · Python, LLM security, prompt injection, AES-GCM
Researched LLM vulnerabilities with proof-of-concepts, mitigated prompt injection with ML-based guardrails, and built envelope encryption (KMS/DEK/KEK) for RAG databases.
Distributed Minesweeper — Fault-Tolerant SMR · Python, distributed systems, TCP sockets, Docker
Fault-tolerant multiplayer game on a replicated state machine (SMR) with Leader–Follower topology and two-phase commit.
Atlas IoT IDE & Smart-Space Orchestration · Python, JavaScript, WebSockets, UDP/TCP, IoT
Web-based visual IDE that discovers IoT devices over UDP multicast and programs them through a node-based interface.
Performance-Oriented Ray Tracing · C++23, Intel TBB, performance, SoA/AoS
Optimised a 3D ray tracer in C++23 with Intel TBB, comparing memory-layout strategies (SoA vs. AoS).
Heuristic Search (A*) · Python, A*, admissible heuristics, AI
Route planner for a spy plane using A* and admissible heuristics over a cost map from Gaussian radar models.
More projects — hybrid cryptography, CSP solvers, operating-systems concurrency, linear-programming optimisation and computer vision — on github.com/nico-maire and the interactive portfolio nico-maire.github.io.

Education & certifications

B.Sc. in Computer Science (Bilingual Programme) *2023 – Present*
Universidad Carlos III de Madrid (UC3M)

- **Erasmus+ exchange:** Università di Bologna (UniBO), Italy — Feb–Jun 2026.
- **International mobility:** UADE (Buenos Aires, Argentina) — Jul–Dec 2026.
- Santander Grant for academic record · Cambridge C1 Advanced (English).